Practice Quiz 08C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 08

Surjectivity and composition; a converse counterexample.

1. Let $f : A \to B$ and $g : B \to C$ be functions.

   (a) Prove that if $g \circ f$ is onto, then g is onto.

   Solution Suppose $g \circ f$ is onto. Let $c \in C$ be arbitrary. There is an $a \in A$ with $(g \circ f)(a) = c$. Set $b = f(a)$, which belongs to B. Then $g(b) = g(f(a)) = c$. Every element of C therefore has a preimage under g, so g is onto. ∎

   (b) Give finite sets A, B, C and functions $f : A \to B$, $g : B \to C$ for which g is onto but $g \circ f$ is not onto. Verify both claims.

   Solution Take $A = \{0\}$ and $B = C = \{0, 1\}$. Define $f(0) = 0$, $g(0) = 0$, and $g(1) = 1$. The function g is onto because 0 and 1 are preimages of the corresponding elements of C.

   The composite has only one input and satisfies $(g \circ f)(0) = 0$. Thus $1 \in C$ has no preimage under $g \circ f$, so the composite is not onto. ∎

Solutions revised using GPT-6 Astra in the Codex harness.